Catching Up to Global NGO Regulatory Change: A Responsible AI Workflow for Comparative Legal Coding

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Abstract

Comparative research on civil society regulation is constrained by the scale, complexity, and multilingual nature of legal texts. Rules affecting civil society organizations (CSO) and nonprofit organizations (NGO) are rarely contained in a single statute; they are distributed across civil codes, tax laws, registration laws, reporting requirements, and regulations. This paper develops and evaluates a responsible AI workflow for scaling comparative legal coding. The workflow uses large language models (LLMs) as research assistants and validates their outputs against expert-coded data and the legal texts themselves. Using a multilingual corpus of civil society laws in English, French, Chinese, Russian, and Arabic, we compare two coding pipelines: a full-law first pass followed by retrieval-based verification, and a retrieval-only one-pass pipeline. Each pipeline is tested with GPT-5 and assessed for agreement with human coders, precision, recall, evidence faithfulness, and cost and runtime. Both pipelines reach about 73 percent vote-weighted accuracy against human coders. Verification does not change accuracy, but it raises evidence faithfulness from 0.69 to 0.95, making nonzero codes traceable to the statute. The retrieval-only pipeline costs less ($0.14 versus $0.60 per source file) but misses rules distributed across a law. The main contribution is a responsible AI workflow for comparative legal research and an error typology for diagnosing recurring errors in LLM-assisted legal coding for NGO regulations.

Keywords: large language models, responsible AI, comparative legal coding, civil society regulation, legal NLP, NGO law

1 Introduction

Given the disagreements among scholars in comparing NGO regulation, can AI, through large language models (LLMs) improve our prospects for building better datasets? Despite the number of academic projects to assess NGO regulations (Breen, Dunn and Sidel, 2017, 2019; Swiney, 2019; Dupuy, Ron and Prakash, 2016; Dupuy, Fransen and Prakash, 2021; Chaudhry, 2022; Chaudhry and Heiss, 2022; Christensen and Weinstein, 2013; DeMattee, 2019*a*,*b*; DeMattee and Swiney, 2021) there is little agreement on where, when, and how regulations are becoming restrictive or if they even are. The number of types of nonprofit, nongovernmental organizations and regulations, the speed of change, and the lack of agreement on which laws restrict versus regulate organizations and disagreement on whether these are implemented correctly or not all complicate the task.

In this paper we examine the potential for using large language models to assist with coding NGO regulations, including civil codes, tax regulations, registration and reporting laws. We test accuracy across 48 laws in five different languages (English, French, Chinese, Russian, Arabic) with detailed performance and errors examined closely in five cases (Kenya, Mozambique, PRC, Russia, and Egypt). Previous projects have focused on narrow subsets of regulations for example constitutional provisions on freedom of association or expression (Bakke, Mitchell and Smidt, 2020; Christensen and Weinstein, 2013) or financial restrictions (Dupuy, Ron and Prakash, 2016; Chaudhry and Heiss, 2018; Bromley, Schofer and Longhofer, 2020; Oelberger and Shachter, 2021) or selective countries (Bloodgood, Tremblay-Boire and Prakash, 2014; DeMattee, 2023*a*,*b*), regions (DeMattee, 2020), or regime types (Heiss, 2017; Toepler et al., 2020; Swiney, 2019). Constitutional protections of freedoms of association, assembly, and expression are vital layers of law, but these measures need implementation in national regulations to support civil society (Rutzen, 2015; Biekart, Kontinen and Millstein, 2023). Governments' increasing use of economic measures to silence civil society groups by limiting funding (Dupuy, Ron and Prakash, 2016; Dupuy and Prakash, 2018; Oelberger and Shachter, 2021; Toepler et al., 2020; Bromley, Schofer and Longhofer, 2020) and the location of restrictions on protest in public security laws suggests a broader set of regulations is relevant. Without the full legal context for all NGO identities, including their operation as foundations, nonprofit corporations, nonprofit associations, and social enterprises, it is difficult to assess what is opening (or closing), and how, and thus for practitioners to advocate for better regulations to protect civil society space.

Generative AI enables the rapid coding and recoding of regulations according to alternative designs to test contrasting interpretations quickly and consistently. In this paper we test a protocol to use GPT-5 to parse laws into provisions and then test these provisions against an established matrix for determining if a provision is restrictive or permissive. To evaluate the protocol, we measure vote-weighted accuracy, precision, recall, evidence faithfulness, error types, cost, and runtime by pipeline stage, regulatory category, and language, against expert human coding of the same laws with the CSO Regulatory Regime Matrix (DeMattee, 2019*b*), and we interpret the results in light of disagreement among existing NGO-law datasets (Dupuy and Prakash, 2018; Bromley, Schofer and Longhofer, 2020; Chaudhry, 2022).

This project answers calls for a more systematic analytic approach to understanding civil society's legal framework. By enhancing the ability to quickly parse and code laws according to multiple schema we can resolve academic debates about the state of laws across countries and organization types and move on to substantive concerns with how to address issues of NGO repression where it occurs. Coding comparative NGO regulation more easily but accurately will enable scholars to resolve academic debates which might be based on misunderstandings in order to generate better recommendations about how governments, NGOs, and foundations might be able to change laws to enable and protect civil society. This method of legal classification should also work on case law, expanding understanding of the legal environments confronting CSOs.

2 Previous Work

2.1 The challenges of LLMs for legal coding

Benchmarks establish a shared vocabulary to evaluate tasks LLM models can and cannot perform based on prior tests of LLMs performance, but benchmarks do not directly translate to comparative legal coding. Recent work finds that LLMs can be sensitive to slight prompt alterations, which is especially important in legal tasks, as these models are only partially aligned with human legal judgments (Guha et al., 2023; Purushothama et al., 2025). Choi (2024) finds that GPT-4 classifies U.S. Supreme Court opinions about as well as human coders, with limited gains from specialized prompting or fine-tuning for that task. Our setting differs, multilingual statutes coded against a 56-provision regulatory matrix, and we evaluate not only label agreement but also whether the cited evidence is grounded in the legal text.

We are evaluating whether LLMs can serve as assistants in comparative legal coding, not resolve legal disputes regarding interpretation or provide legal reasoning as judges would. Thus, legal benchmarks give us a good starting point for model selection, but we are focused on multiple capabilities of LLMs, where legal reasoning is one of multiple necessities for reliable legal coding. We task them with identifying relevant legal provisions, extracting their institutional statements, and classifying them into a framework with text-grounded evidence and a justification for the coding. This is a complex task, as errors in legal coding can lead to misleading comparative claims about regulatory regimes and the civic space.

2.1.1 Retrieval, evidence faithfulness and hallucinations

One of the biggest risks in legal applications of LLMs is persuasive but unsupported claims about what the law says, which can come from generations of false or misleading outputs, referred to as hallucinations. Dahl et al. (2024) develop a typology of legal hallucinations and show that LLMs can generate inconsistent outputs with respect to legal facts and struggle to identify hallucinations. Retrieval methods may help dampen this risk but do not fully eliminate it, as Magesh et al. (2025) find. We examine this with two pipelines to compare the tradeoffs of retrieval systems. Thus, we question not whether retrieval 'solves' hallucination, but when and where it fails, and whether there is a systematic way to identify them.

A responsible legal coding workflow must base its correct final label in faithful evidence from relevant legal text. Work on retrieval-augmented attribution, including RARR, develops frameworks for checking whether model-generated outputs are correctly attributed and for separating supported claims from unsupported ones (Gao et al., 2023). Other past research similarly evaluates self-critique as a vital part of the retrieval-augmented generation (RAG) process, and emphasizes that systems should be evaluated on relevance, accuracy and faithfulness (Asai et al., 2024; Yu et al., 2025).

2.1.2 LLMs as annotators

Computational social science work on LLMs as annotators has emphasized that model-generated labels should be treated as a first-pass measurement rather than ground truth, with human intervention remaining central for validation (Ziems et al., 2024; Egami et al., 2023). Pangakis and Wolken (2025) find that their performance as annotators varies significantly across tasks, and argue that evaluation must be based on human-generated labels.

2.1.3 Multilingual capabilities

Comparative civil society research is inherently multilingual. Laws affecting CSOs are written in many languages, and translation can introduce drift even before AI enters the workflow. We review a multilingual corpus of laws to compare the performance of different languages on this specific task. Yong et al. (2025) find a significant language gap in LLM safety research; even in high-resource languages, performance differs from English. For this reason, multilingual evaluation is part of this project's robustness checks, as it is essential for its use as a coding tool, where translation drift has been a problem long before generative AI. English-language files serve as the main benchmark, while French, Chinese, Russian, and Arabic files are used to diagnose robustness and failure modes.

3 Methodology

This design evaluates whether GPT-5 can support comparative legal coding of civil society regulation when embedded in a human-in-the-loop workflow. First, legal coding should be evaluated against human judgments without pretending that human coders always agree. Second, model outputs should be evaluated beyond solely assigning the same label as human coders and also by whether they provide traceable statutory evidence for their claims.

3.1 Corpus

The evaluation uses 67 source files corresponding to 48 civil society laws, some laws appear in more than one language. We treat the operational unit of model evaluation as the source file, while the substantive legal corpus contains 48 distinct laws. The corpus includes English-language files as the main benchmark and French, Chinese, Russian, and Arabic files as robustness checks.

3.2 CSO Regulatory Regime Matrix

The coding instrument used in this project is the CSO Regulatory Regime Matrix developed by DeMattee (2019*b*). The matrix makes possible comparative study of civil society regulation by decomposing regulatory regimes into specific legal provisions.

It has two organizing logics. First, provisions are grouped by the part of the regulatory regime they govern: *Governance*, *Formation*, *Operations*, and *Resources*.

- Governance provisions concern the creation, authority, oversight, and accountability of regulatory bodies.
- Formation provisions concern the legal status of CSOs and the procedures through which organizations register, incorporate, continue, or dissolve.
- Operations provisions concern what organizations may or must do in pursuit of their activities, including licensing, permits, reporting, and restrictions on particular activities.
- Resources provisions concern financial and non-financial assets, including foreign funding, taxation, fundraising, property, audits, and financial reporting.

Second, provisions can be understood along a restrictive–permissive continuum. Some provisions increase burdens, limit autonomy, or create coercive state authority, while others provide legal status, accountability, remedies, or rights.

The matrix expresses each provision using Institutional Grammar, using the ADICO syntax (Crawford and Ostrom, 1995). Institutional Grammar translates natural-language legal rules into structured institutional statements. **A** refers to the **actor** or attribute to whom the rule applies; **D** is the **deontic** operator, such as may, must, or must not; **I** is the **aim**, action, or legal effect; **C** identifies the **condition** under which the rule applies; and **O**, or "**or else**," identifies the consequence of compliance or noncompliance.

Because not all laws express all elements of a legal rule in one sentence, a legally equivalent rule may be distributed across multiple provisions. For this reason, we instruct the model to read each matrix item as a structured legal relationship and to ask whether the statute regulates the same relationship or produces the same legal effect, even if the law does not mirror the matrix wording.

Each source file (law) is coded against the 56 provisions of the CSO Regulatory Regime Matrix. Each model output receives one of three labels:

- **1**: the statute supports, permits, implements, or legally enables the matrix provision or a legally equivalent rule;
- **0**: the statute is silent, ambiguous, too generic, only indirectly related, or does not provide legally important support for the matrix provision;

- **-1**: the statute negates, prohibits, restricts, reverses, or removes the matrix provision or a legally equivalent rule.

The LLM task is therefore not ordinary topic classification. Although we take inspiration from the growing field, this is a rather more complex legal measurement task: for each matrix provision, the model must decide whether the source law contains, omits, or negates a legally equivalent institutional statement, and it must support that decision with traceable statutory evidence.

3.3 Human benchmark and vote-weighted evaluation

We preserve the distribution of human coder labels, as legal coding can involve genuine ambiguity. Each law was coded by one to three expert human coders. We use vote-weighted accuracy as the primary accuracy metric. For each law-provision pair, the model receives credit equal to the share of human coders who selected the same label. A model prediction of (1) against human labels ([1,1,0]) receives (2/3) credit; a prediction of (0) receives (1/3) credit; and a prediction of (-1) receives no credit. Overall vote-weighted accuracy is the total number of matching human votes divided by the total number of human votes.

In addition to accuracy, we compute vote-weighted precision, recall, and F1. Nonzero recall captures how much legally relevant content the model finds, and nonzero precision captures how often the model's nonzero claims are supported by human-coded labels.

3.4 AI workflow

We evaluate two pipelines and three model stages. Pipeline A is a two-stage full-law and verification workflow. Pipeline B is a retrieval-only baseline.

3.4.1 A1: full-law first-round coding

In the first stage of Pipeline A, GPT-5 receives the full extracted legal text and codes all 56 matrix provisions. This stage is a broad first-pass legal coder. It has access to the entire law, which allows it to identify rules that may be distributed across different parts of a statute.

For each provision, the model returns a label (1/0/-1), statutory evidence, and a brief explanation. Evidence is required to be a verbatim quotation from the legal text, and explanations are required to

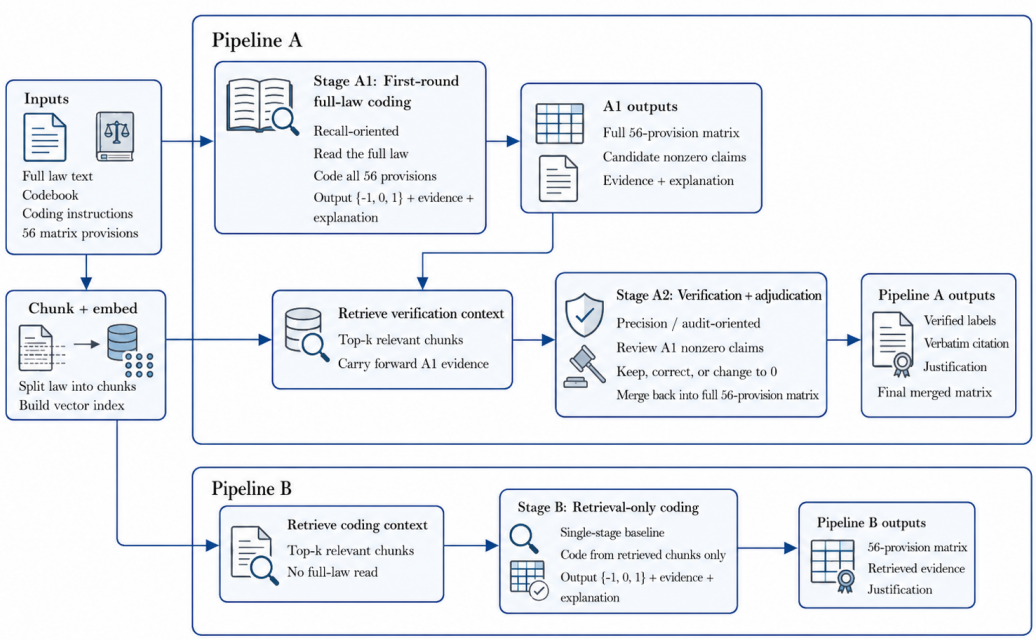


Figure 1: Legal coding workflows evaluated. Pipeline A separates first-pass full law coding (A1) from verification (A2), while pipeline B is a retrieval-only baseline.

distinguish the legal reasoning from the evidence itself.

3.4.2 A2: verification and adjudication

The second stage of Pipeline A verifies the nonzero claims generated by A1. For each A1 nonzero output (1/-1), the system retrieves relevant chunks from the law and asks GPT-5 to adjudicate whether the first-round label is supported by the provided statutory text. The verifier prompt treats the first-round output as a provisional legal coding judgment.

The purpose of A2 is to serve as an audit layer. It tests whether nonzero claims can be grounded in legal text and produces a final label with evidence that is more directly tied to the text available during verification.

3.4.3 B: retrieval-only baseline

Pipeline B is a retrieval-only coding baseline. Instead of reading the full law, the system first retrieves chunks relevant to the matrix and then asks GPT-5 to code all 56 provisions using only the retrieved text. We design this to evaluate whether restricting the model to selected passages improves evidence grounding and/or reduces overreach because of prompt saturation. It tests a common assumption in

retrieval-augmented generation (RAG), that retrieval will necessarily improve legal reliability. In NGO regulation, that is spread across bills, tax codes, and other documents, retrieval may reduce unsupported claims but it may also miss legally relevant rules.

3.5 Retrieval and chunking

The retrieval system divides each source file into token-based chunks with overlap. We build an index using embeddings and retrieve chunks based on matrix provisions and evidence hints. For Pipeline A verification, retrieval is targeted to the A1 nonzero claim being verified. For Pipeline B, retrieval is matrix-oriented and supplies the model with a restricted set of chunks for coding the entire matrix.

All runs used GPT-5 (`gpt-5`) with default reasoning settings and JSON-schema-constrained outputs, run in June 2026, with one run per source file and pipeline. Each source file was split into 512-token chunks with a 64-token overlap and embedded with `text-embedding-3-small`; chunks were ranked by cosine similarity. In A2, the query for each nonzero A1 claim combined the matrix provision with the A1 evidence, and the verifier received the top three chunks together with their neighbouring chunks. In Pipeline B, one broad query and one query per matrix provision each retrieved their top two chunks, and the deduplicated set was passed to the model.

To measure evidence faithfulness, each evidence field was split into quoted units of at least 20 characters. An evidence row counts as grounded when at least one unit appears in the stage context, either verbatim or after normalizing quotation marks, dashes, whitespace, and letter case; no fuzzy-matching threshold was applied. Grounding therefore checks whether cited text can be located, not whether the coding inference is correct, and extraction or encoding differences can mark a real quotation as ungrounded.

3.6 Evidence faithfulness

For every model stage, we evaluate evidence faithfulness separately from label performance. Evidence faithfulness measures whether the evidence cited by the model is grounded in the text that was available to that stage. For A1, the relevant context is the full source file. For A2, the relevant context is the retrieved verification text. For B, the relevant context is the retrieved chunks used by the retrieval-only baseline.

A legally useful model output must be more than legally plausible, it must be auditable. Evidence

faithfulness asks whether a human reviewer can locate the cited support in the relevant legal text. It does not by itself prove that the label is correct. A model may cite real statutory text and still draw the wrong coding inference. Conversely, a model may assign a label that matches human coders while providing weak or ungrounded evidence.

3.7 Prompt development framework

For comparative legal coding, prompt development should be iterative and grounded in observed reasoning errors. We present the following prompt development framework. The general procedure is applicable to other comparative legal coding efforts in distinct fields using large language models. The iterations performed for this project are reported in Appendix A.

1. **Translate the coding typology into rules.** The prompt should begin from a clear coding instrument with decision rules that the model is expected to apply. It should explain how the model should handle edge cases observed by human coders in previous efforts. In this project, the coding instrument was a 56-provision CSO Regulatory Regime Matrix using three labels: `1`, `0`, and `-1`. This process can be initiated with a LLM itself, asking to draft the initial prompt accompanied by the coding typology, example source files that will be coded and what an output shall look like.

2. **Run on a small pilot subset.** Before scaling, the prompt should be tested on a small and diverse set of laws. This will expose recurring model behaviors that would make the coding unreliable or difficult to audit.

3. **Inspect reasoning and evidence for systematic errors.** Human review is essential in this step. Matching labels are important, but pilot review should mainly examine the model's explanations and cited evidence. Prompt changes should respond to recurring patterns. In this project, the main pilot errors included over-deletion in the verification stage, rigid ADICO matching, and rigid exact-word matching.

4. **Revise the prompt to correct the observed patterns.** Each prompt revision should correspond to a specific observed failure mode. For example, if the model misses functionally equivalent statutory language, the prompt should clarify legal equivalence. If the model overreaches from generic language, the prompt should clarify when to code `0`.

5. **Freeze the prompt before final evaluation.** Once the prompt has been revised based on the pilot, it should be fixed before running any final evaluation. Later qualitative review can be used to interpret errors, but not to alter the evaluated outputs.

4 Findings

The results show that the three evaluated stages serve different functions. The full-law first pass is the strongest classifier, especially for recovering nonzero legal content. The verification stage substantially improves evidence faithfulness. The retrieval-only baseline is competitive in overall accuracy but weaker at recovering nonzero provisions.

4.1 Overall classification performance

Table 1 reports vote-weighted classification metrics for the full corpus of 67 source files. A1, the full-law first-round stage, has the strongest classification performance overall. It achieves the highest vote-weighted accuracy, nonzero recall, nonzero F1, macro F1, and weighted F1. A2, the verification stage, has slightly lower classification performance than A1, especially in nonzero recall. B, the retrieval-only baseline, is close to A2 in vote-weighted accuracy but has the lowest nonzero recall and nonzero F1.

Table 1: Overall vote-weighted classification performance

Stage	Accuracy	Precision	Recall	F1	Macro F1	Weighted F1
A1 first round	0.7332	0.5896	0.6871	0.6346	0.4842	0.7313
A2 final	0.7280	0.5845	0.6203	0.6019	0.4672	0.7217
B retrieval-only	0.7289	0.5837	0.5704	0.5770	0.4602	0.7179

Note: Precision, recall, and F1 refer to vote-weighted nonzero classification. Accuracy is vote-weighted accuracy against the distribution of human-coded labels.

The most important pattern is not the small difference in overall accuracy, but the larger difference in recall. A1 recovers more of the nonzero legal content identified by human coders. This is consistent with the design of the stage: because A1 reads the full legal text, it can identify rules that are distributed across sections or expressed through definitions, procedures, and institutional powers. By contrast, A2 and B are constrained by retrieval. This constraint improves auditability, but it can also remove legal context needed to identify a provision.

The retrieval-only pipeline is therefore useful but not dominant. Its overall accuracy is close to A1

and A2, but its lower nonzero recall suggests that retrieval alone may miss relevant legal content. For comparative civil society research, this matters because the most substantively important task is often identifying the relevant provisions first.

4.2 Evidence faithfulness

Table 2 reports evidence faithfulness by stage. Evidence faithfulness measures whether the model's cited evidence is grounded in the text available to that stage. A2 clearly performs best on this dimension. Its evidence faithfulness is 0.9483, compared with 0.6927 for A1 and 0.7446 for B.

Table 2: Evidence faithfulness by stage

Stage	Rows	Evidence rows	Faithfulness
A1 first round	3,752	1,227	0.6927
A2 final	3,752	1,103	0.9483
B retrieval-only	3,752	1,010	0.7446

Note: Evidence faithfulness is the share of evidence rows grounded in the stage context. The relevant context is the full law for A1, retrieved verification context for A2, and retrieved chunks for B.

4.3 Accuracy and Auditability

We estimate OLS models comparing the final output of the two-step pipeline A (round A2) with the retrieval-only pipeline B baseline. These models are presented to test whether the descriptive pipeline comparison remains stable after accounting for differences across regulatory categories, source-file languages, laws, and matrix provisions.

The regression sample excludes A1 because A2 depends on A1 and represents the final output of Pipeline A. The sample also excludes Arabic files, which have text normalization issues. The retained sample includes English, French, Chinese, and Russian source files. Standard errors are clustered by law/source file.

Table 3 reports six models. Models 1–3 use row-level vote-weighted accuracy as the outcome. Models 4–6 use evidence faithfulness as the outcome, measured among rows where the model provided evidence. The key predictor is an indicator for Pipeline A final output, where Pipeline B is the reference category. The control models include category and language controls, with Governance as the reference category and English as the reference language. The fixed-effects models include law and provision fixed effects.

Table 3: Comparison of accuracy and evidence faithfulness

	(1) Accuracy pipeline only	(2) Accuracy + controls	(3) Accuracy + FE	(4) Faithfulness pipeline only	(5) Faithfulness + controls	(6) Faithfulness + FE
Pipeline A final (A2)	0.001	0.001	0.001	0.184***	0.181***	0.182***
	(0.006)	(0.006)	(0.007)	(0.029)	(0.028)	(0.029)
Formation		0.068***			-0.020	
		(0.023)			(0.022)	
Operations		0.105***			0.001	
		(0.022)			(0.021)	
Resources		0.186***			-0.040	
		(0.024)			(0.024)	
Chinese		0.009			-0.167**	
		(0.063)			(0.082)	
French		0.088			-0.035	
		(0.066)			(0.068)	
Russian		-0.000			0.007	
		(0.082)			(0.018)	
Category controls	No	Yes	No	No	Yes	No
Language controls	No	Yes	No	No	Yes	No
Law fixed effects	No	No	Yes	No	No	Yes
Provision fixed effects	No	No	Yes	No	No	Yes
Clustered SE by law/source file	Yes	Yes	Yes	Yes	Yes	Yes
N	6,832	6,832	6,832	1,864	1,864	1,864
R-squared	0.000	0.028	0.266	0.079	0.107	0.235
Adjusted R-squared	-0.000	0.027	0.254	0.078	0.104	0.186
AIC	8256.2	8077.6	6369.3	974.0	927.7	851.5
BIC	8269.9	8132.2	7168.3	985.0	972.0	1482.0

Notes: Models with standard errors clustered by law/source file. The accuracy outcome is row-level vote-weighted accuracy. The evidence-faithfulness outcome is coded among rows where the model provided evidence. Pipeline B retrieval-only is the reference pipeline. Governance is the reference category and English is the reference language. Fixed-effects models include law and provision fixed effects. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The results show no meaningful difference in vote-weighted accuracy between pipeline A final and Pipeline B. The coefficient for pipeline A final is approximately 0.001 across all accuracy specifications. This means that A2 final outputs are not statistically distinguishable from retrieval-only outputs in label accuracy. However, the evidence faithfulness models show that pipeline A final is associated with an 18 point increase in evidence faithfulness relative to Pipeline B, and this estimate remains stable after adding controls and law/provision fixed effects.

These results lead us back to our initial assumptions. A2 is not valuable primarily because it improves classification accuracy, it comes from producing auditable outputs by grounding cited evidence more reliably in the legal text. In other words, the verification stage improves the actual research usefulness of the output even when it does not materially change aggregate label accuracy.

The category controls also confirm that Governance is the most difficult matrix category. Relative to Governance, Formation, Operations, and Resources all have higher vote-weighted accuracy. Some of the remaining model difficulty is driven less by pipeline choice than by the legal complexity of the provisions being coded.

Table 4 reports interaction models, varying the pipeline A final coefficient by matrix category. They

show that the main accuracy result is largely unchanged, pipeline A final does not show a broad accuracy advantage over pipeline B. The only category-specific accuracy difference appears in Formation, where the pipeline A final interaction is positive. The evidence-faithfulness model again shows a more consistent auditability advantage. Pipeline A has a positive and statistically significant faithfulness coefficient in the reference category, Governance. The Formation interaction is also positive and statistically significant, suggesting an additional faithfulness advantage in that category.

Table 4: Pipeline-by-category interaction models

	(A) Accuracy Pipeline × Category	**(B) Evidence faithfulness Pipeline × Category**
Pipeline A final (A2)	-0.009	0.132***
	(0.011)	(0.036)
Formation	0.050**	-0.057
	(0.025)	(0.036)
Operations	0.102***	-0.023
	(0.023)	(0.045)
Resources	0.187***	-0.084*
	(0.025)	(0.050)
Pipeline A final × Formation	0.035**	0.071**
	(0.015)	(0.036)
Pipeline A final × Operations	0.007	0.047
	(0.016)	(0.052)
Pipeline A final × Resources	-0.001	0.086
	(0.014)	(0.054)
Chinese	0.009	-0.167**
	(0.063)	(0.082)
French	0.088	-0.035
	(0.066)	(0.068)
Russian	-0.000	0.007
	(0.082)	(0.018)
Category controls	Yes	Yes
Language controls	Yes	Yes
Law fixed effects	No	No
Provision fixed effects	No	No
Clustered SE by law/source file	Yes	Yes
N	6,832	1,864
R-squared	0.028	0.110
Adjusted R-squared	0.026	0.105
AIC	8081.6	928.5
BIC	8156.7	989.3

Notes: OLS models with standard errors clustered by law/source file. The accuracy outcome is row-level vote-weighted accuracy. The evidence-faithfulness outcome is coded among rows where the model provided evidence. Pipeline B retrieval-only and Governance are the reference categories. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Altogether, the final output (round A2) of pipeline A is not meaningfully more accurate than the

retrieval-only, one-round baseline (B), but it is substantially more faithful to the legal text provided. For comparative legal coding, the value of the verification layer lies in making model outputs reliable, and thus auditable.

4.4 Performance by regulatory category

Table 5 reports vote-weighted classification metrics by matrix category. A1 has the highest nonzero F1 in all four categories. The category-level results also show substantial variation in difficulty. Formation is the strongest category by nonzero F1, while Governance is the weakest.

Table 5: Vote-weighted classification performance by category

Category	Stage	Accuracy	Precision	Recall	F1
Formation	A1 first round	0.7292	0.7891	0.7557	0.7720
Formation	A2 final	0.7099	0.7790	0.6775	0.7247
Formation	B retrieval-only	0.6932	0.7663	0.5928	0.6685
Governance	A1 first round	0.6246	0.4000	0.7152	0.5131
Governance	A2 final	0.6326	0.4024	0.6623	0.5006
Governance	B retrieval-only	0.6442	0.4082	0.6258	0.4941
Operations	A1 first round	0.7429	0.5614	0.6230	0.5906
Operations	A2 final	0.7398	0.5561	0.5829	0.5692
Operations	B retrieval-only	0.7398	0.5504	0.5401	0.5452
Resources	A1 first round	0.8098	0.5812	0.6156	0.5979
Resources	A2 final	0.8049	0.5777	0.5296	0.5526
Resources	B retrieval-only	0.8141	0.6050	0.5188	0.5586

Note: Precision, recall, and F1 refer to vote-weighted nonzero classification.

4.5 Language robustness

The English-language files are the main benchmark. Table 6 summarizes the main language-level results. In English, all three stages have similar vote-weighted accuracy, but A2 has substantially higher evidence faithfulness.

The language results should be interpreted cautiously because the non-English samples are small. They are most useful for identifying failure modes. Arabic is the clearest example. A1 has very high label metrics in Arabic, but extremely low evidence faithfulness. The labels may align with human codes, while the cited evidence fails to match the extracted or stage-level text. This issue may reflect OCR or text extraction.

Table 6: Language-level robustness checks

Language	Stage	Files	Accuracy	Precision	Recall	Faithfulness
English	A1 first round	46	0.7252	0.5720	0.6752	0.7934
English	A2 final	46	0.7255	0.5770	0.6364	0.9699
English	B retrieval-only	46	0.7263	0.5797	0.6144	0.7983
French	A1 first round	6	0.7738	0.6047	0.7429	0.6782
French	A2 final	6	0.7738	0.6000	0.6857	1.0000
French	B retrieval-only	6	0.7708	0.4839	0.2143	0.8710
Chinese	A1 first round	6	0.7381	0.6316	0.6429	0.6122
Chinese	A2 final	6	0.7381	0.6306	0.6250	0.8947
Chinese	B retrieval-only	6	0.7143	0.5857	0.7321	0.5714
Russian	A1 first round	4	0.7031	0.5088	0.4957	0.7368
Russian	A2 final	4	0.7031	0.5000	0.4615	0.9815
Russian	B retrieval-only	4	0.7299	0.5800	0.4957	0.8200
Arabic	A1 first round	3	0.8869	0.8919	0.9340	0.0360
Arabic	A2 final	3	0.6369	0.9412	0.4528	0.6275
Arabic	B retrieval-only	3	0.5833	0.9302	0.3774	0.0698

Note: Precision and recall refer to vote-weighted nonzero classification.

4.6 Error Typology

To interpret the quantitative results, we manually reviewed selected model outputs from five substantive cases: Egypt's 2017 Law No. 70 on Associations and Civil Work, Kenya's 2007 Political Parties Act, Russia's 2012 Foreign Agents Law, China's 2016 Overseas NGO Law, and Mozambique's 1991 Law on the Right of Association. The full qualitative review is reported in the appendix.

The review suggests that disagreement with the human benchmark is not primarily a matter of simple hallucination. In many cases, the model cites real language and provides a legally plausible explanation, but the human benchmark codes the provision differently, where there is disagreement on whether a provision is legally equivalent to the matrix provision being coded.

Table 7 summarizes the typology. The false-negative cell captures cases where the human benchmark identifies a nonzero provision but the model codes zero; the false-positive cell captures cases where the model identifies a nonzero provision but the human benchmark codes zero.

A fifth pattern cuts across the four cells: **Human–AI disagreement**. In these cases, both the model and the human have legally understandable readings, but they apply different thresholds for legal equivalence, domain scope, or code polarity (1/-1). These disagreements are difficult to resolve, given the complexities of comparative legal coding on whether legal language should count as legally equivalent to a typology provision.

Table 7: Qualitative error typology in confusion-matrix form

True positive (TP) Agreement on presence of the provision.	**False negative (FN)** **FN1:** Misaligned evidence or retrieval miss. **FN2:** Underclassification despite grounded evidence.
False positive (FP) **FP1:** Unsupported or ungrounded evidence. **FP2:** Overinference/incorrect reasoning despite grounded evidence.	**True negative (TN)** Agreement on absence of the provision.

4.7 Cost and runtime

In order for the workflow to be a useful tool for comparative legal coding, it is important to ensure it is scalable in terms of cost and runtime. Because large language models are typically priced by usage rather than by a fixed fee, scalability depends partly on how much text the workflow sends to and receives from the model.

This usage is measured in *tokens*, small units of text used by the model to process language. Tokens are roughly comparable to word pieces, a short word may be one token, while a longer word, punctuation, or formatting may be split into multiple tokens. Input tokens refer to the legal text, matrix provisions, retrieved excerpts from the laws, and instructions sent to the model. Output tokens refer to the codes, evidence, and explanations generated by the model.

Latency is measured as cumulative model-call runtime, cost is estimated using GPT-5 API prices of $1.25 per million input tokens and $10.00 per million output tokens (OpenAI, 2026).

Table 8: Token expenditure, latency, and estimated cost by model stage

Pipeline stage	Calls	Latency hours	Avg. lat. min/file	Input tokens	Output tokens	Total tokens	Cost/file
A1: full-law first pass	67	3.23	2.89	1,492,065	756,932	2,248,997	$0.14
A2: retrieval verification	1,216	8.81	7.89	5,120,878	2,437,276	7,558,154	$0.46
B: Retrieval-only	67	2.25	2.01	673,457	861,742	1,535,199	$0.14

Table 8 separates the cost of the full-law first pass from the cost of the verification layer. The first stage of Pipeline A is relatively inexpensive: the full-law pass costs approximately $0.14 per source file, similar to the retrieval-only baseline. The main cost difference comes from A2, the retrieval-based verification stage. Because A2 verifies nonzero first-round claims separately, it required 1,216 logged model calls and added approximately $0.46 per source file. In total, Pipeline A costs approximately $0.60 per source file, while Pipeline B costs approximately $0.14 per source file.

This difference is expected given the design of the workflows. A1 uses a single large full-law call to identify candidate provisions. A2 then makes additional targeted calls to verify the evidence and final label for nonzero claims. Pipeline B is much cheaper because it restricts the model to retrieved context and makes only one call per source file. In practical terms, coding one source file costs roughly fourteen cents with either the A1-only or B-only approach, but approximately sixty cents with the full A1+A2 workflow.

The cost difference should be interpreted alongside the classification and evidence-faithfulness results. Pipeline B is faster and less expensive, but it is not consistently superior as a classifier and is more vulnerable to retrieval misses. A1 is comparatively inexpensive and provides the strongest first-pass classification performance. A2 adds cost, but it produces the most audit-ready workflow by improving evidence faithfulness and making nonzero claims easier to inspect. For research settings where human review and traceable evidence are essential, the additional cost of the verification layer may be justified.

5 Discussion

The human benchmark is not an infallible measure of legal truth. While we find that both AI pipelines reach roughly 73 percent vote-weighted accuracy in coding NGO laws as compared to the human-coded benchmark (DeMattee, 2019*b*), this result is better than it seems on face value. Human coders have also been found to have low degrees of agreement in categorizing NGO laws as restrictive or not. In a comparison of nine research projects which code NGO regulations (Christensen and Weinstein, 2013; Bakke, Mitchell and Smidt, 2020; Dupuy, Ron and Prakash, 2016; Bromley, Schofer and Longhofer, 2020; Oelberger and Shachter, 2021; Bloodgood, Tremblay-Boire and Prakash, 2014; Swiney, 2019), only three pairs of projects coding which countries are restrictive or not are correlated at the 70 percent level or higher. Of the possible 45 combinations, only Dupuy et al. and Bromley et al. (77 percent), Dupuy et al. and Oelberger and Shachter (91 percent), and Bromley et al. and Oelberger and Shachter (78 percent) largely agree on the distribution of restrictive regulation across countries. Given Bromley et al. and Oelberger and Shachter explicitly use Dupuy et al.'s publicly available data to build their own datasets, this is not surprising. Among the other possible pairs of projects, the correlation coefficients range from -0.19 to 0.52. This strongly suggests both that coding NGO laws is a difficult task and that the LLMs are not performing worse than humans despite not reaching our hoped for degree of accuracy

(90 percent or higher). The LLM coding tools have the added benefit of providing researchers both a rationale for the coding and the specific evidence extracted directly from the legal text. Researchers can then confirm (or change) the coding for themselves quickly, without needing to sift through hundreds of pages of law, and with confidence, given the amount of information provided alongside the -1, 0, or 1.

LLM NGO law coding tools provide an additional benefit beyond human coding which is that the rules used to categorize legal provisions as restrictive, neutral, or enabling can be easily changed mid-project without the loss of hundreds of hours of time (not to mention researchers' morale and the project's momentum). Should repeated patterns of gaps between the LLM coding and previous datasets emerge, suggesting that an alternative definition of restrictiveness is more appropriate or useful for a project, then the LLM tool can be quickly altered and the coding repeated with limited loss of time, effort, or funding. For example, the spread of foreign interference regulations across Western countries in the past several years (Bolleyer et al., 2025) has generally concerned political parties and explicitly partisan political actors. These laws have thus been omitted from many researchers' analyses of NGO laws (e.g. Chaudhry, 2022) which exclude political parties from their definition of NGO and thus their analysis. In many countries, however, political parties are nonprofit organizations (although not charities) and thus do fit within the classic definition of NGO as organizations without official government representations and a strict non-distribution of revenue constraint (Martens, 2002; Weisbrod, 1988; Salamon, Sokolowski and Haddock, 2017; Bassi, Aquino Alves and Cordery, 2025). The addition of new foreign interference regulation to the relevant legal text would thus mean a number of countries currently coded as neutral or enabling are actually restrictive. In a related example, it might be more productive for a researcher to code the restrictiveness of legal provisions by the category of organization (charity/public benefit organization, foundation, community/grassroots association, nonprofit corporation, and social enterprise) rather than category of activity. The LLM tools presented here could be easily and quickly adapted to this task.

6 Limitations and Future Work

This study has several considerations that should guide interpretation of the results. First, the evaluation is based on 67 source files corresponding to 48 distinct laws. English-language files make up the main benchmark, while French, Chinese, Russian, and Arabic files are best understood as robustness checks

and diagnostic cases, where all non-English files have an English translated counterpart in the 48-law English corpus.

Second, an output may be marked as weakly grounded because of extraction or formatting limitations, not hallucinated evidence. This distinction is especially important for OCR-heavy and non-Latin source files. Arabic was the clearest example, classification performance and evidence faithfulness diverged sharply, suggesting that source-language extraction and evidence normalization require additional audit before making strong claims.

Finally, the three evaluated stages have different limitations. A1 is useful as a broad full-law first pass and generally provides stronger nonzero recall, but its evidence is less consistently grounded. A2 substantially improves evidence traceability, but it can still produce wrong-but-grounded outputs when it maps real legal text to the wrong matrix item. B is cheaper and more constrained, but retrieval-only coding can miss rules that are distributed across articles and sections of the same law. No single stage should be treated as a complete substitute for human review.

Future work can extend the same evaluation to additional models and repeated runs to measure model differences and output stability.

Legal AI systems can fail at multiple levels, whether it is retrieval, reasoning, or translation. The contribution of the workflow is to make these risks visible. By producing structured labels, evidence, explanations, and diagnostic metrics, the system helps human researchers decide where to trust, what to inspect, or which legal codes to revise.

7 Conclusion

We test GPT-5 as an assistant for comparative legal coding of civil society regulation. The answer is yes, it can code, but that does not mean it is a task humans should leave for the model to do independently. The results show that when its outputs are structured, grounded in legal text, and include reasoning that an expert can evaluate, they can be effective as first-pass coders, helping humans scale significantly the amount of documents processed and inevitably form part of our analyses.

When evaluating a language model as a coder, accuracy and auditability are equally important. A1, the full-law first pass, performs best as a broad classifier and recovers the most legal provisions that a human would. A2, the verification stage, does not substantially improve label accuracy, but it

significantly improves evidence faithfulness. Pipeline B is faster and cheaper, but its reliance on retrieved context makes it more vulnerable to missing legal rules distributed across articles.

Our error typology shows what kinds of disagreements human coders can expect and makes them easier to audit. Many of them are not simple hallucinations; in some cases, the model cites real legal text and provides a plausible explanation, but the human benchmark rejects the legal mapping. These cases turn on legal equivalence and domain scope. It is for humans to decide the level of specificity required for a rule to count as support for a typology provision.

The workflow developed here shows how LLMs can help researchers catch up to global NGO regulatory change without leaving the interpretive work involved in legal coding. LLMs are not lawyers or policy analysts; humans are. We will inevitably disagree, though it is by figuring out the best way for humans and AI to collaborate that we can build the comparable data needed to understand the rapidly changing and contracting liberties of civil society.

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A Prompt Development

A.1 Pilot Iteration

In this project, we ran a five-law pilot to identify recurring patterns that would have led to a high disagreement rate from our human benchmark.

A.1.1 A2 Was Too Deletion-Heavy

The largest issue in the pilot was that the verification stage, A2, was too willing to turn A1's nonzero labels into `0`. A2 often treated A1's output as a claim to disprove. This made A2 too conservative on substantive labels.

The recurring pattern was that A1 identified plausible legal content, while A2 rejected it because the statute did not use the exact wording of the matrix provision. This showed that the verification prompt needed to distinguish between unsupported claims and legally supported claims expressed in different language.

A.1.2 The Model Treated ADICO Too Rigidly

The matrix provisions follow Institutional Grammar, but the pilot showed that the model sometimes treated each ADICO element as a required checklist. It would reject a provision if the statute did not explicitly contain the exact actor, deontic, aim, condition, and consequence in one place.

That standard was too strict for comparative legal coding. Statutory rules are often distributed across definitions and exceptions. A legal rule can be present even if all elements do not appear in a single sentence or in the same order as the matrix provision.

A.1.3 The Model Demanded Exact Wording Instead of Legal Equivalence

The pilot also showed that the model sometimes missed functionally equivalent rules. For example, if the matrix stated that an organization *may register*, while the statute stated that an organization *shall apply for registration*, a rigid reading could treat the two as different because the deontic language differs. In legal terms, however, the statute may still create the same registration pathway or legal capacity.

Similar problems arose when the model looked for literal matrix terms such as "negligent," "overstepping," "before commencement," "after commencement," or "or else," even when the statute created the

same legal effect using different wording. This led us to prioritize legal effect over textual similarity.

A.1.4 The Model Sometimes Overreached from Broad or Adjacent Language

The pilot also showed the opposite risk. In some cases, the model treated broad administrative authority, background institutional language, or adjacent legal provisions as enough to support a specific matrix item. This meant the final prompt needed to balance a dynamic where functionally equivalent legal rules should count if they meet the legal relationship that the IG statement makes, but generic, speculative, or merely adjacent legal language should not.

If the prompt was too strict, the model would reject provisions that did not mirror the matrix wording. If it was too loose, the model would accept broad or vaguely related language as sufficient support. The final prompt was designed to sit between the extremes, flexible enough to recognize legal equivalence, but strict enough to reject generic, speculative, or merely adjacent language.

A.2 How the Prompt Was Revised

The final prompt revisions responded directly to the systematic errors observed in the pilot.

A.2.1 Functional Legal Equivalence Was Added

We instruct the model to code legal and functional equivalents, instead of requiring exact wording from the matrix.

> Use those elements to understand what is being regulated, but coding should be functional and cohesive. Decide whether the statutory text regulates the same legal relationship or produces the same legal effect as the matrix provision.

A.2.2 ADICO Was Reframed as Guidance, Not a Checklist

The final prompt instructs the model to use ADICO to understand the legal relationship being coded, but not to require every ADICO element to appear in one sentence or in the same wording.

> Do not require every ADICO element to appear in one sentence or in the same form as the matrix provision. Consider the statutory rule cohesively across related clauses.

A.2.3 The Prompt Clarified When to Code `0`

To prevent overreach, the final prompt also made the `0` category more explicit.

> Code 0 when support is merely speculative, generic, or based on broad regulatory authority.

A.2.4 A2 Was Reframed from Adversarial Deletion to Verification

The A2 prompt was revised so that it no longer treated A1's nonzero labels as automatically suspicious.

> Treat the first-round label as a provisional legal coding judgment. It is not automatically correct, but it is also not automatically suspect.

A.3 Lessons from prompt iteration

Even after prompt revisions, disagreement will remain. The methodological contribution of the prompt-development process is to make model outputs more auditable. By identifying recurring reasoning problems, coders can learn what kinds of outputs require special human scrutiny.

The prompt development framework is a step toward an auditable legal measurement process where the model proposes and justifies codes, while human reviewers know where to look, what errors to expect, and how to evaluate whether the cited evidence actually supports the legal coding decision.

B Error Typology

We manually review model outputs for five cases

- Egypt's 2017 Law No. 70 on Associations and Civil Work
- Kenya's 2007 Political Parties Act
- Russia's 2012 Foreign Agents Law
- China's 2016 Overseas NGO Law
- Mozambique's 1991 Law on the Right of Association

For each law, we inspect the model code, human labels, cited evidence, explanation, and retrieved context where available. The cases were selected because they vary in legal domain, language, model performance, and type of disagreement. Egypt provides a high-performing association-law case; Kenya tests the boundary between political party regulation and CSO regulation; Russia and China test restrictive foreign-agent or foreign-NGO regimes in both English and original-language versions; and Mozambique provides a cleaner association-law case centered on broad legal rights and regulatory delegation.

The most important finding is that model disagreement with the human benchmark is not primarily a matter of simple hallucination. In many cases, the model cites real statutory language and gives a legally plausible explanation, but the human benchmark codes the provision differently. As such, evidence faithfulness is necessary for auditability, but it is not sufficient for legal correctness.

Table B.1: Qualitative error typology

Type	**Error or disagreement pattern**
FN: retrieval miss	Relevant text exists in the law, but retrieval did not surface enough of it for the model to support a nonzero code.
FN: underclassification	The model sees the general legal issue, but treats the evidence as too indirect or incomplete to satisfy the matrix provision.
FP: unsupported or ungrounded evidence	The model gives a nonzero code, but the cited evidence is missing, poorly grounded, mistranscribed, or not traceable to the available stage context.
FP: grounded overinference or incorrect reasoning	The model cites real statutory language, but maps it too broadly or incorrectly to the matrix provision.
Human–AI disagreement	Both the model and human benchmark have legally understandable readings, but they apply different thresholds for legal equivalence, domain scope, or code sign (1/-1).

1. False negatives caused by retrieval limitations

The first type of false negative occurs when the relevant statutory text exists in the law, but the retrieval-based stage does not surface enough of that text for the model to code the provision correctly. This is especially important for Pipeline B and for the A2 verification stage, because both rely on retrieved chunks. When the relevant provision is absent from the retrieved context, the model may correctly refuse to make a nonzero claim even though the full statute supports one.

This problem appears in the retrieval-only coding of Egypt's 2017 Associations Law for Governance Provision 7.

Matrix provision coded:

> GOVERNANCE PROVISION 7. [Agency] [may] [publish certain information about a CSO, or seize CSO property and documents] [with or without reasonable cause after law's commencement] [or else choose other tactics to receive information from CSOs].

Human benchmark: `1`

Model code: `0` in the retrieval-only stage.

In the full-law first round, the model correctly identified statutory support for this provision. The relevant legal text cited by the model was:

> Article 10: "The Administrative Entity undertakes to enter the association and take necessary actions to publish the summarized articles of association of the association on the official website of the Administrative Entity..."
>
> Article 27: "Representatives of the Administrative Entity... may enter the premises... to monitor its activities and review its records..."

The full-law model explained:

> "The administrative authority is empowered and required to publish association summaries and has authority to enter and review records; this satisfies the permissive capability to publish information or access documents."

However, the retrieval-only stage coded the same provision as zero, with no supporting evidence. Its explanation was:

> "The law authorizes inspections and record reviews, but the provided text does not show publication authority by the agency or seizure powers."

The retrieval-only model appears to have recognized that inspection language existed, but it did not have enough context to connect that inspection language to the separate publication authority in Article 10. The law contains the relevant rule, and the full-law stage can identify it, but the stage operating under retrieved context does not see enough of the law to support a nonzero code.

This pattern is important because it shows that retrieval can improve auditability while also creating a new failure mode. The verifier or retrieval-only model may be cautious and evidence-disciplined, but if the retrieved context is incomplete, that caution can become undercoding.

2. False negatives caused by underclassification

A second type of false negative occurs when the model sees the general legal issue, but classifies the law as too indirect, too generic, or insufficiently tied to the precise matrix provision. In these cases, the model is not missing the topic entirely. Instead, it applies a narrower interpretation of legal equivalence than the human benchmark.

Russia's 2012 Foreign Agents Law provides a useful example for Governance Provision 2.

Matrix provision coded:

> GOVERNANCE PROVISION 2. [Agency] [must] [explain penalty for particular offenses, or explain a 'general penalty' for offenses where no penalty is expressly provided] [before commencement of the law] [or else it is negligent in its duties].

Human benchmark: `[1, 1]`

Model code: `0`

Because the model coded the provision as absent, it did not provide statutory evidence. Its explanation was:

> "Penalties are established directly in the Criminal Code (e.g., Article 330.1), but the statute does not require the agency to explain penalties or issue a general penalty rule."

The disagreement here is about the threshold of specificity required by the matrix. The human coders treated the law's penalty structure as sufficient to support the provision. The model, however, required a more explicit agency duty to explain penalties or issue a general penalty rule. The model is therefore not missing the existence of penalties. It recognizes that penalties exist, but concludes that they are not assigned to the agency in the particular way described by the matrix.

This is underclassification. The model sees a related legal mechanism but declines to classify it as equivalent to the matrix item. In a comparative coding task, this kind of conservative legal mapping can reduce recall by turning substantively relevant provisions into zeros.

Mozambique's 1991 Law on the Right of Association provides a similar example.

Matrix provision coded:

> GOVERNANCE PROVISION 2. [Agency] [must] [explain penalty for particular offenses, or explain a 'general penalty' for offenses where no penalty is expressly provided] [before commencement of the law] [or else it is negligent in its duties].

Human benchmark: `[1, 1]`

Model code: `0`

The model explained:

> "The law contains penalties/sanctions in general terms, but there is no provision requiring an agency to explain penalties for particular offenses or a general penalty rule before commencement."

Again, the disagreement is not that the model failed to notice the existence of legal sanctions. Rather, the model refused to infer an agency duty from a more general penalty framework. These cases show that underclassification can result from the model applying a stricter evidentiary threshold than the human benchmark, especially for matrix provisions that describe administrative duties in more specific terms than the statute itself uses.

3. False positives caused by unsupported or ungrounded evidence

A third category consists of false positives where the model codes a nonzero provision but the supporting evidence is missing, weakly grounded, mistranscribed, or not traceable enough to the available stage context. These are the closest cases to what is often called hallucination. However, in this project many such cases are better described as evidence-support failures rather than wholly fabricated legal claims. The model may be referring to real parts of the law, but its citation does not ground cleanly in the stage context or does not support the specific matrix provision as stated.

Egypt's 2017 Associations Law provides an example for Governance Provision 8.

Matrix provision coded:

> GOVERNANCE PROVISION 8. [Government and agency] [must] [create or empower self-regulation forum for CSO self-governance] [after law's commencement] [or else it is negligent in its duties].

Human benchmark: `0`

Model code: `1`

Grounding status: ungrounded.

The model cited:

> Article 82: "Associations and national foundations... shall establish one regional federation among them... A regional federation has the following functions..."

> Article 84–85: Establish specialized federations and a General Federation with defined self-governance functions.

The model explained:

> "The law mandates and empowers CSO federations (regional, specialized, general) as forums for collective self-governance and coordination."

The human benchmark coded the provision as zero. This example has two distinct issues. First, the evidence was not grounded cleanly in the stage context. Second, even if the cited federation provisions are real, the model appears to overstate their legal significance. A federation that coordinates, represents, trains, or supports associations is not necessarily a self-regulatory forum in the sense required by the matrix. The model's output therefore combines weak evidence grounding with a broad inference from adjacent institutional language.

This category is useful because it separates unsupported evidence from more subtle grounded over-inference. In this case, the citation itself does not pass the grounding check, which makes the output less auditable. Even if a human reviewer can later locate related federation provisions elsewhere in the law, the model's specific evidence trail is not strong enough to support the code as presented.

4. False positives caused by grounded overinference or incorrect reasoning

The fourth category is the most substantively important. In these cases, the model cites real statutory language and the evidence is grounded, but the legal mapping to the matrix provision is too broad or inconsistent with the human benchmark. These are not simple hallucinations. They are cases where the model finds real law but reasons too expansively from it.

Kenya's 2007 Political Parties Act provides a clear example. The model repeatedly treated political-party regulation as functionally equivalent to CSO regulation. For Formation Provision 1, the model coded the provision as present.

Matrix provision coded:

> FORMATION PROVISION 1. [CSOs] [must not] [operate as informal, voluntary associations] and instead must register with the government [or else face penalty for non-compliance].

Human benchmark: `[0, 0]`

Model code: `1`

Grounding status: grounded.

The statutory text cited by the model was:

> Section 18(1): "An association of persons or an organization shall not operate or function as a political party unless it has first been registered in accordance with the provisions of this Act."

The model explained:

> "Operating as a political party requires registration."

The statutory text is real and the model's explanation is internally coherent. The problem is domain scope. The matrix provision concerns CSOs, NGOs, associations, foundations, and related nonprofit legal forms. The Kenyan statute regulates political parties. The model treats political parties as association-like organizations and therefore maps the registration rule onto the CSO matrix. The human benchmark does not. This is a grounded false positive caused by overinference: the model correctly identifies a registration requirement, but applies it to a legal domain that the human coders treat as outside the target CSO regulatory regime.

A similar pattern appears in the same law for Formation Provision 13.

Matrix provision coded:

> FORMATION PROVISION 13. [CSOs] [must] [pay processing/application fee] [in addition to an application that must be completed as prescribed] [or else the registration or reregistration application will not be reviewed].

Human benchmark: `[0, 0]`

Model code: `1`

The model cited:

> Section 18(2)(d): "be accompanied by the prescribed fee"; Reg. 3(2): "issue a certificate... on payment by the applicant of the fees specified in the Second Schedule"; Second Schedule: "Application for provisional registration... 100,000; Application for full registration... 500,000."

The model explained:

> "Applications require prescribed fee payment for processing and issuance of certificates."

Again, the evidence is real. The error is not that the model fabricated a fee. The error is that it mapped a political-party registration fee onto a CSO registration-fee provision. This makes Kenya especially useful for identifying overinference from legally adjacent but substantively different domains.

5. Human–AI interpretive disagreement

The fifth category is broader than a conventional error type. Some disagreements are not easily described as the model being obviously wrong or the human benchmark being obviously self-evident. Instead, both the human and model interpretations are legally understandable, and the disagreement reflects the difficulty of applying a general comparative matrix to complex statutory language. These cases are especially important because they reveal not only model limitations, but also the interpretive burden carried by the coding scheme itself.

Mozambique's 1991 Law on the Right of Association illustrates this kind of disagreement for Governance Provision 5.

Matrix provision coded:

> GOVERNANCE PROVISION 5. [Government] [may] [create or empower cabinet minister to make new rules regulating CSOs on certain matters] [after law's commencement] [or else leave the regulatory regime unchanged].

Human benchmark: `[0, 0]`

Model code: `1`

Grounding status: grounded.

The model cited:

> Article/Clause 20: "It is the responsibility of the Council of Ministers to regulate this law."

The model explained:

> "The Council of Ministers (cabinet) is empowered to issue regulations governing associations."

This is a difficult case. The model's reasoning is plausible: the Council of Ministers is a cabinet-level body, and the statute expressly assigns it responsibility for regulating the law. However, the human benchmark codes the provision as zero. The likely disagreement is over the level of specificity required

by the matrix. The human coding appears to require a more specific delegation of rulemaking authority over CSO matters, while the model treats a general regulatory clause as functionally equivalent.

The same law produces a disagreement in coding direction for Governance Provision 7. Here the human coders and the model both treat the provision as legally relevant, but assign it opposite signs.

Matrix provision coded:

> GOVERNANCE PROVISION 7. [Agency] [may] [publish certain information about a CSO, or seize CSO property and documents] [with or without reasonable cause after law's commencement] [or else choose other tactics to receive information from CSOs].

Human benchmark: `[-1, -1]`

Model code: `1`

The model cited:

> Article 5(3): "The recognition is in the form of an order published in the Boletim da Republica."
>
> Article 5(4): "The statutes of recognized associations are also published in the Boletim da Republica."
>
> Article 12(3): "The declaration of public utility is published in the Boletim da Republica."

The model explained:

> "The law authorizes publication of recognition orders, statutes, and public-utility declarations concerning associations."

The model cites real legal text, and its evidence is grounded. The disagreement concerns whether publication of recognition orders and statutes is equivalent to an agency power to publish information about CSOs in the sense contemplated by the matrix. The model reads formal publication requirements as supporting the provision, while the human coders code the law as negating it. Unlike a false positive, this is not a disagreement over whether the provision is present, but over its direction (`1` versus `-1`). The case shows that grounded evidence can still support a disputed legal mapping: the model may be equating formal legal publication with a discretionary agency power, or the human coding may reflect a reading of the wider regime that the cited articles alone do not show.

Russia's 2012 Foreign Agents Law shows a similar interpretive boundary problem for Governance Provision 1.

Matrix provision coded:

GOVERNANCE PROVISION 1. [Government] [must] [create or appoint agency to implement and enforce the law] [at law's commencement] [or else it is negligent in its duties].

Human benchmark: `[-1, -1]`

Model code: `1`

Grounding status: grounded.

The model cited:

> Article 2, 3) in Article 13.1, clause 10: "Information contained in the documents of a noncommercial organization performing the functions of a foreign agent ... shall make up the register of noncommercial organizations performing the functions of foreign agents, the keeping of which is the responsibility of the authorized body. The procedure of keeping the said register shall be established by the authorized body."
>
> Article 2, Article 32, clause 3: "Noncommercial organizations ... shall submit to the authorized body documents ..."

The model explained:

> "The law designates an 'authorized body' to maintain the register, set procedures, receive reports, and supervise CSOs, thereby appointing an agency to implement and enforce the regime at commencement."

The model's reading is legally coherent: the law assigns responsibilities to an authorized body. The human benchmark, however, codes the provision as `-1`. This suggests that the disagreement is not only about whether an implementing body exists, but also about the legal effect captured by the matrix code. The model reads institutional assignment as support for the provision. The human benchmark appears to treat the foreign-agent regime as negating or restricting the relevant governance relationship. This type of disagreement highlights the interpretive difficulty of the task and shows why human review remains necessary even when evidence is grounded.

China's 2016 Overseas NGO Law provides another example of this broader interpretive problem. For Governance Provision 1, the model identifies the Ministry of Public Security and provincial public security organs as responsible bodies for registration and supervision.

Matrix provision coded:

GOVERNANCE PROVISION 1. [Government] [must] [create or appoint agency to implement and enforce the law] [at law's commencement] [or else it is negligent in its duties].

Human benchmark: `-1`

Model code: `1`

The model cited statutory language assigning public security authorities responsibility for registration, annual inspection, supervision, and enforcement. Its explanation was that the law assigns implementation and enforcement authority to designated government bodies. This is a reasonable reading of the administrative structure of the statute. Yet the human benchmark codes the provision differently. As with Russia, the disagreement likely turns on how the matrix code should capture restrictive regulatory architecture: the model sees an implementing agency, while the human coding may interpret the broader legal relationship as restrictive or inconsistent with the provision's positive formulation.

These examples show that human expertise remains essential for resolving domain boundaries, code polarity, and the level of specificity required for a legal rule to count as support for a comparative legal provision.

C Prompts

C.1 A1

```
# ROLE
You are a legal coding assistant for comparative civil society regulation.

# TASK
Code the provided statute against the CSO Regulatory Regime Matrix. Return one code for each
↪ of the 56 matrix provisions.

# HOW TO APPLY THE MATRIX
Each matrix provision describes a legal rule about civil society regulation. The bracketed
↪ ADICO elements help identify the relevant actor, legal operator (deontic), action or
↪ legal effect (aim), condition, and consequence (or else).
```

```
Use those elements to understand what is being regulated, but coding should be functional and
↪ cohesive. Decide whether the statutory text regulates the same legal relationship or
↪ produces the same legal effect as the matrix provision.
A law may support a provision through direct or legally equivalent wording, definitions,
↪ procedures, institutional powers, or related clauses that together establish the relevant
↪ rule.
Do not require exact mirroring of the matrix grammar. At the same time, do not stretch broad
↪ or adjacent legal language to fit a provision that the law does not actually support.

# CODES
Use only:
1 = The statute directly, legally, or functionally supports implementation of the matrix rule
↪ or a legally equivalent version of it.
0 = The statute is silent, ambiguous, too generic, only indirectly related, or missing
↪ legally important support for the matrix rule.
-1 = The statute directly, legally, or functionally supports negation, prohibition,
↪ restriction, removal, or reversal of the matrix rule.

# CODING GUIDANCE
- Code 1 only when the statute provides clear legal support for the specific matrix
↪ provision.
- Code 0 when support is merely speculative, generic, or based on broad regulatory authority.
- Code -1 only when the statute clearly reverses, prohibits, or removes the matrix rule.
- Do not require exact wording from the matrix. Legal and functional equivalents count.
- Do not require every ADICO element to appear in one sentence or in the same form as the
↪ matrix provision. Consider the statutory rule cohesively across related clauses.
- A stronger mandatory rule may satisfy a permissive may provision when it creates the same
↪ legal capacity. But a mere permission usually does not satisfy a mandatory must provision
↪ unless the surrounding text creates the required duty or consequence.
- Definitions, exceptions, penalties, procedures, and cross-references may be used when they
↪ directly support the code.
- Do not rely on outside knowledge.

# CSO SCOPE
```

```
Treat CSOs as including charities, foundations, nonprofit organizations, NGOs, civil
↪ associations, voluntary associations, public-benefit organizations, and similar legal
↪ forms regardless of local terminology.

# EVIDENCE
- Evidence must be verbatim quotation(s) from the statute.
- Include article, section, or clause references when present.
- Evidence must quote the statute in the statute's original language. Explanation must be in
↪ English, even when the statute is in another language.
- Use null when the statute is silent or no specific evidence supports the code.
- Put reasoning in Explanation, not in Evidence.

# EXPLANATION
Explain briefly why the evidence supports the code. For Code 0, explain what is missing,
↪ ambiguous, too generic, or only indirectly related.
```

C.2 A2

```
# ROLE
You are a legal coding verifier for comparative civil society regulation.

# TASK
Verify whether a first-round non-zero CSO matrix code is supported by the law.

# VERIFICATION POSTURE
Treat the first-round label as a provisional legal coding judgment. It is not automatically
↪ correct, but it is also not automatically suspect.
Use only the provided statutory text as support. Do not rely on outside knowledge.
Do not require exact matrix wording. Legal and functional equivalents count.

# HOW TO APPLY THE MATRIX
Each matrix provision describes a legal rule about civil society regulation. The bracketed
↪ ADICO elements help identify the relevant actor, legal operator (deontic), action or
↪ legal effect (aim), condition, and consequence (or else).
```

```
Use those elements to understand what is being regulated, but coding should be functional and
↪ cohesive. Decide whether the statutory text regulates the same legal relationship or
↪ produces the same legal effect as the matrix provision.
A statute may support a provision through direct or legally equivalent wording, definitions,
↪ procedures, institutional powers, or related clauses that together establish the relevant
↪ rule.
Do not require exact mirroring of the matrix grammar. At the same time, do not stretch broad
↪ or adjacent legal language to fit a provision that the statute does not actually support.

# CODES
Return:
1 = The statute directly, legally, or functionally supports implementation of the matrix rule
↪ or a legally equivalent version of it.
0 = The statute contains no direct, legal, or functional support for the matrix rule.
-1 = The statute directly, legally, or functionally supports negation, prohibition,
↪ restriction, removal, or reversal of the matrix rule.

# VERIFICATION RULES
- Prefer the best legally supported final code over mechanical deletion.
- Keep or correct to a non-zero label when the law functionally regulates the same legal
↪ relationship or produces the same legal effect as the matrix provision.
- Change a non-zero first-round label to 0 only when the law does not support the specific
↪ matrix provision, is merely generic or adjacent, or regulates a materially different
↪ legal issue.
- Do not require exact wording from the matrix, but do require legal equivalence.
- Do not require every ADICO element to appear in one sentence or in the same form as the
↪ matrix provision. Consider the statutory rule cohesively across related clauses.
- Do not change a non-zero label to 0 merely because the matrix's or else consequence is not
↪ stated verbatim, if the statute provides an ordinary legal consequence such as refusal,
↪ denial, ineligibility, suspension, cancellation, revocation, fine, penalty, written
↪ reasons, appeal, judicial review, or mandatory correction.
- If the statute itself creates, defines, empowers, or regulates the relevant institution or
↪ legal category, do not zero the claim solely because the duty is enacted directly by
↪ statute rather than separately performed by an agency.
```

```
- If a permissive matrix provision says an actor may do something, a stronger mandatory
↪  version may still support code 1 when it creates the same legal capacity, forum, right,
↪  or mechanism.

# EVIDENCE
- Evidence must be verbatim quotation(s) from the provided chunks.
- Include article, section, or clause references when present.
- Evidence must quote the statute in the statute's original language. Explanation must be in
↪  English, even when the statute is in another language.
- Use null if the correct label is 0 and no on-point quotation supports a non-zero code.
- Put reasoning in Explanation, not in Evidence.

# EXPLANATION
Explain briefly whether the provided evidence supports the first-round claim. If correcting
↪  to 0, explain what is missing, ambiguous, too generic, or only indirectly related.
```

C.3 B

```
# ROLE
You are a legal coding assistant for comparative civil society regulation.

# TASK
Code the provided law for all 56 CSO Regulatory Regime Matrix provisions.

# HOW TO APPLY THE MATRIX
Each matrix provision describes a legal rule about civil society regulation. The bracketed
↪  ADICO elements help identify the relevant actor, legal operator (deontic), action or
↪  legal effect (aim), condition, and consequence (or else).
Use those elements to understand what is being regulated, but coding should be functional and
↪  cohesive. Decide whether the provided statutory text regulates the same legal
↪  relationship or produces the same legal effect as the matrix provision.
The law may support a provision through direct or legally equivalent wording, definitions,
↪  procedures, institutional powers, or related clauses that together establish the relevant
↪  rule.
```

```
Do not require exact mirroring of the matrix grammar. At the same time, do not stretch broad
↪ or adjacent legal language to fit a provision that the provided chunks do not actually
↪ support.

# CODES
Use only:
1 = The law directly, legally, or functionally supports implementation of the matrix rule or
↪ a legally equivalent version of it.
0 = The law contains no direct, legal, or functional support for the matrix rule.
-1 = The law directly, legally, or functionally supports negation, prohibition, restriction,
↪ removal, or reversal of the matrix rule.

# CODING GUIDANCE
- Code 1 only when the statute provides clear legal support for the specific matrix
↪ provision.
- Code 0 when support is merely speculative, generic, or based on broad regulatory authority.
- Code -1 only when the statute clearly reverses, prohibits, or removes the matrix rule.
- Do not require exact wording from the matrix. Legal and functional equivalents count.
- Do not require every ADICO element to appear in one sentence or in the same form as the
↪ matrix provision. Consider the statutory rule cohesively across related clauses.
- A stronger mandatory rule may satisfy a permissive may provision when it creates the same
↪ legal capacity. But a mere permission usually does not satisfy a mandatory must provision
↪ unless the surrounding text creates the required duty or consequence.
- Definitions, exceptions, penalties, procedures, and cross-references may be used when they
↪ directly support the code.
- Do not rely on outside knowledge.

# CSO SCOPE
Treat CSOs as including charities, foundations, nonprofit organizations, NGOs, civil
↪ associations, voluntary associations, public-benefit organizations, and similar legal
↪ forms regardless of local terminology.

# EVIDENCE AND CHUNK IDS
- Evidence must be verbatim quotation(s) from the provided chunks.
```

- Include article, section, or clause references when present.
- Evidence must quote the statute in the statute's original language. Explanation must be in English, even when the statute is in another language.
- Every non-zero code must include supporting ChunkIds.
- Use null evidence and an empty ChunkIds list when the provided chunks do not support a non-zero code.
- Put reasoning in Explanation, not in Evidence.

EXPLANATION

Explain briefly why the provided evidence supports the code. For Code 0, explain what is missing, ambiguous, too generic, or only indirectly related in the provided chunks.